

Software Requirements Specification (SRS)

Hand Gesture Based Virtual Mouse Web Application

1. Introduction

1.1 Purpose

This document specifies the requirements for the Hand Gesture Based Virtual Mouse Web Application.

The system enables users to control mouse operations using hand gestures captured via webcam.

1.2 Scope

The application allows real-time hand gesture recognition using MediaPipe and Python Flask.

Mouse actions such as cursor movement, click, scroll, and drag are performed without physical contact.

1.3 Definitions and Abbreviations

SRS – Software Requirements Specification

HCI – Human Computer Interaction

API – Application Programming Interface

FPS – Frames Per Second

2. Overall Description

2.1 Product Perspective

The system is a web-based application that uses a client webcam and server-side processing.

2.2 Product Functions

- Capture webcam input
- Detect hand landmarks
- Identify gestures
- Perform mouse actions
- Display live feedback

2.3 User Classes

- General User
- Developer
- Researcher

2.4 Operating Environment

- Web Browser
- Webcam
- Python Flask Server
- Cloud Deployment

2.5 Constraints

- Webcam permission required
- Internet dependency
- Limited server resources

2.6 Assumptions

- Stable network
- Functional camera
- Supported browser

3. System Requirements

3.1 Functional Requirements

FR1: Capture live video feed

FR2: Detect hands using MediaPipe

FR3: Track finger movements

FR4: Recognize gestures

FR5: Execute mouse actions

3.2 Non-Functional Requirements

Performance: Low latency

Reliability: Continuous operation

Security: No video storage

Usability: Simple gestures

Scalability: Multi-user support

4. External Interface Requirements

User Interface:

Web-based graphical interface with live video.

Hardware Interface:

Webcam and display device.

Software Interface:

MediaPipe, Flask, OpenCV.

Communication Interface:

HTTP / HTTPS protocol.

5. System Architecture

Backend:

Flask server with MediaPipe and OpenCV.

Frontend:

HTML, CSS, JavaScript.

Deployment:

Hosted on cloud using Render with Gunicorn.

6. Implementation Details

Backend Implementation:

- Flask app creation
- Webcam capture using OpenCV
- Hand landmark detection
- Gesture mapping
- Mouse control using PyAutoGUI

Frontend Implementation:

- Webcam access
- Video streaming
- Gesture feedback

7. Deployment on Render

1. Create requirements.txt and Procfile
2. Push code to GitHub
3. Create Render Web Service
4. Select Python runtime
5. Set Gunicorn start command
6. Deploy and access public URL

8. Future Enhancements

- Custom gesture training

- Multi-hand support
- AI-based improvements
- Mobile compatibility

9. Conclusion

The Hand Gesture Based Virtual Mouse Web Application provides an innovative and touchless way to interact with computer systems using hand gestures and web technologies.